

MA489 Final Report

Application of Recurrent Neural Networks to Radiation Oncology Quality Assurance

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Abstract

This research is motivated by a real-world challenge – before radiation is conducted on a cancer patient, the radiation plan must pass a quality assurance (QA) testing process. It is assumed that the variability in a radiation plan has a large influence on the QA pass rates. Metrics, both from literature and based on the theories of contributors from the Princess Margaret Hospital, measure the variation of a radiation plan. These include, but are not limited to, leaf travel, modulation complexity score, and field span. The statistical significance of the metrics is tested using logistic regression models. Deep learning models such as recurrent neural networks (RNN) have been created to predict QA pass rates and will be analyzed to evaluate the adoptability of machine learning models in the clinical setting. An accurate and interpretable machine learning model is essential for gaining trust from domain users, complying with regulations, and enhancing the decision making of end users. Ideally, the models created through this research can be leveraged by affiliates at the Princess Margaret Hospital and will serve as a framework for integrating machine learning in radiation oncology.

1 Introduction

When a patient is first diagnosed with cancer, it must be decided upon which therapy (or therapies) will be used to treat the cancer. The fundamental idea behind radiation therapy is that tumor cells are more sensitive to radiation than healthy cells. Ideally, the radiation is able to kill or significantly damage the tumor cells. If radiation therapy is pursued, then volumetric modulated arc therapy (VMAT) is a new and popular choice. The focus of my research in radiation therapy is limited to VMAT.

Before radiation can be applied to the patient there are several steps that need to be taken. First, imaging of the location of the tumor must be done and a plan of the dosage must be created. This plan needs to be tested in quality

assurance (QA). It is only after QA, that the patient can receive the radiation therapy.

Radiation therapy is administered to the patient by a linear accelerator (linac). The linac rotates 360° around the treatment bench on which the patient lies. The gantry is the part of the linac which enables the rotation. The linac controls and shapes the radiation beam at various ‘control points’ throughout the treatment. These control points occur at consistent angle intervals of the gantry’s rotation.

The photon beam is shaped by the multi-leaf collimator (MLC). The leaves of a MLC are independently movable metal blades that can be positioned to create a variable opening for the photon beam to pass through, shaping the beam by allowing or blocking specific areas to precisely target the treatment area while minimizing exposure to surrounding healthy tissues. The MLC has two leaf banks. In one of the leaf banks the leaves move from the left side into the isocentre of the photon beam and in the other leaf bank, the leaves move from the right. When the leaf in the left bank and the corresponding leaf in the right bank are at the same location there is no opening between them, meaning no radiation will pass through. Further, the distance between corresponding leaves corresponds to the amount of photon radiation being administered at a specific control point. Also part of the collimator are the jaws. The jaws cover certain areas of the multi-leaf collimator such that the beam cannot pass through these areas. Jaws are used, for example, when the target of the radiation is a smaller location and not all leaves of the collimator need to be exposed. The larger the target tumor, the more leaves and the larger the area left exposed by the collimator jaws. Depending on the machine, a collimator has either two or four jaws. Machines with four jaws have a jaw on the left, right, top, and bottom, whereas ones with two jaws only have a jaw on the top and bottom.

1.1 Treatment Planning

A treatment planner uses the imaging of the tumor to map the dose distribution of the VMAT. This is a large optimization problem in which the treatment planner looks to deliver a homogeneous dose to the tumor while minimizing the radiation that the surrounding organs receive. In Figure 1, there are various shapes that represent the tumor locations, the dose distributions and the surrounding organs. For example the right eye is mapped by a light blue circle. Ideally the optimization would create a 3D continuous dose that is equally distributed at each angle but the delivery of this dosage is constrained by the linac. The motions of the leaves, the gantry angle motion, and the beam intensity delivered by the linac are all constraints that need to be considered in the planning phase. This is a highly complex process and it would be very irresponsible to immediately administer this dosage to the patient. Thus, the plan must undergo quality assurance to ensure that it is safe to be delivered to the patient.

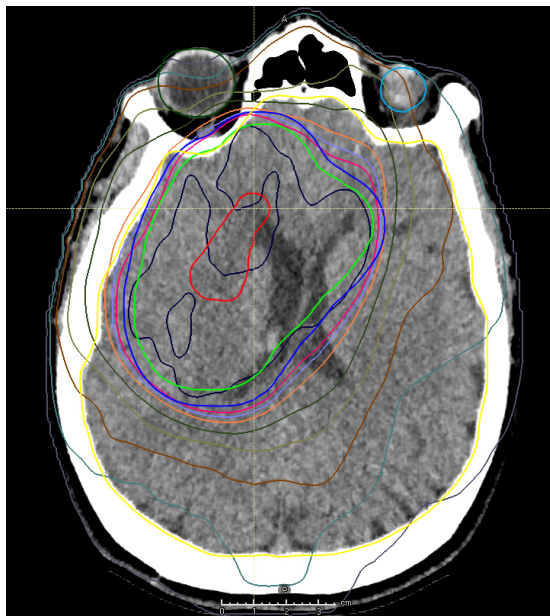


Figure 1: Treatment Planning Shape Example.

1.2 Quality Assurance

In QA, the radiation plan is administered by the linac to a device called an ArcCheck. ArcCheck is a device that contains 1386 detectors which measure the radiation from the linac at various different points in space. Figure 2 shows an ArcCheck device and its detector structure. The success of QA is dependent on the QA pass rate. At each detector, a comparison is made between the measured dosage that the detector receives during QA and the dosage predicted by the treatment planner. If the difference between the two measurements is below a certain threshold then that detector passes the QA. The main pass rate is defined as the percentage of detectors that pass the dose comparison. Whether the patient passes the QA is dependent on the AD pass rate which is the percentage of detectors in the ArcCheck that pass their QA.

QA must be carried out for two reasons:

It is not possible to place detectors in or on the patient during the treatment so there is no way of determining the dosage the patient receives. The measured dose distribution from the QA is used as an approximation of the real dosage that will be received by the patient.

The synergy between the linac and the treatment plan must be evaluated. There are limitations to the motions of the linac's gantry angle and multi-leaf collimator. Testing must be done to determine if the linac can appropriately deliver the treatment plan devised by the planner.

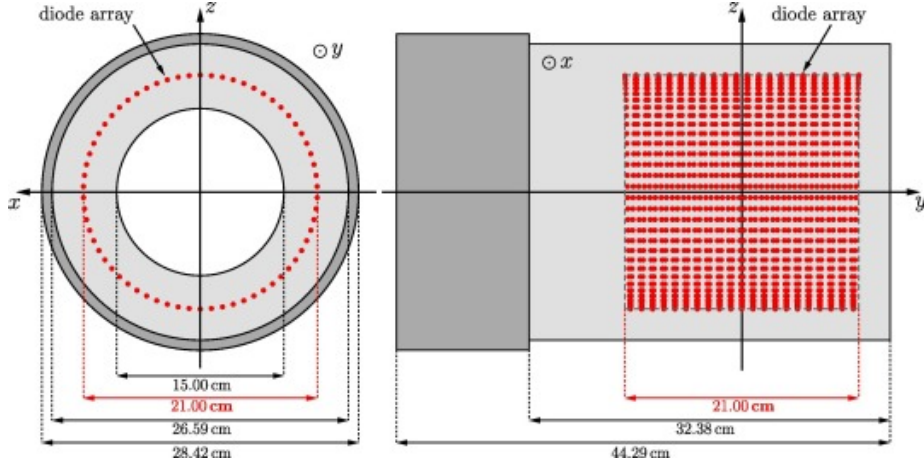


Figure 2: ArcCheck Architecture. [Granville et al.]

2 Related Work

Going into this research, I knew my goal would be to predict the QA pass rate using the data produced regarding the treatment plan of the patient. My first inspiration came from a paper named *Complexity analysis of Volumetric Modulated Arc Therapy prostate plans* by Stathis Kamperis [Kamperis 2021]. In this paper, I discovered complexity metrics and their use as inputs in machine learning models. Complexity metrics are used to extract features of the field through its geometric representation. For example, the area of a field can be calculated by measuring the distance between two adjacent leaf pairs. Ideally these features serve as a representation of the complexity of a treatment field. Although complexity metrics have been used in practice for predicting QA pass rates [Osman et al.], in the article, Kamperis only uses them to cluster test fields into high complexity and low complexity clusters. As mentioned, complexity metrics have been used as predictors in machine learning algorithms such as poisson regression ([Li et al.]), tree-based algorithms such as random forest ([Hirashima et al.], [Ono et al.], [Wall & Fontenot]), SVM ([Granville et al.], [Wall & Fontenot]), and neural networks ([Kimura et al.], [Ono et al.], [Tomori et al.], [Wall & Fontenot], [Wang et al.]). The number of treatment plans used to train these models ranges anywhere from 74 to 1255. Our research has been conducted on 1464 fields from the Princess Margaret Hospital Cancer Clinical Research Unit. I was unable to find any research that employed a Recurrent Neural Network (RNN) algorithm for predicting QA pass rates. I will discuss in the methodology why the RNN algorithm is potentially useful for this specific problem and in the results section, I will evaluate the performance of the RNN model.

3 Methodology

3.1 Data

As mentioned, this research makes use of 1464 treatment fields to train machine learning models. I would like to briefly describe the data representing these fields. The field data is nested into three layers: the treatment field, the control points, and the leaf sets. The treatment field holds information about the entire treatment such as the machine used, the total radiation monitor units (MU), the field's isocentre, and various other attributes. Each field has several control points at which the LINAC can be instructed to change leaf positions. The control point contains information regarding the current gantry angle, the angle of the multi-leaf collimator, the number of leaves, and the current dimensions of the field. Nested within each control point are the leaf sets. As mentioned earlier, there are two left sets; the leaves coming from the right and from the left. Each left set has an equal number of leaves and the leaf positions for each leaf set are stored in an array. The leaf position is the distance of the tip of the leaf from the y-axis. This hierarchical structure is displayed in figure 3. Since every leaf set in the field has the same number of leaves, I have created a matrix representation of the leaf positions of a field. The columns of this matrix represent the gantry angle of the control point and the column is a vector of the leaf positions at that control point. This way, the rows represent the leaf tip positions of a particular leaf. Two matrices are needed to represent the leaf positions of a field, one for each leaf set. Figure 3 also shows the matrix representation of the treatment field.

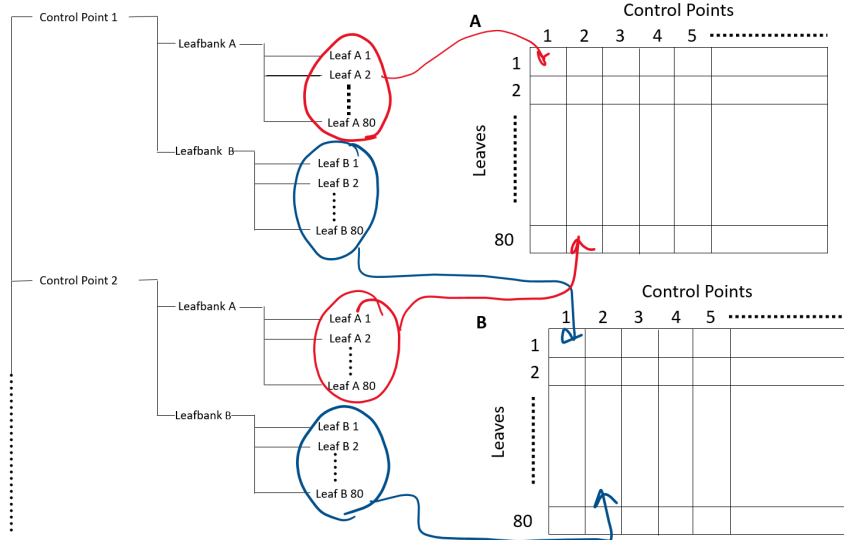


Figure 3: Nested representation of data (left), matrix representation of data (right).

3.2 Complexity Metrics

As discussed in the related work section, I sought out metrics that measure the complexity of a treatment plan. The intended use of these metrics is as predictors variable in the upcoming machine learning models for predicting QA pass rates.

Most of the complexity metrics that I implemented were inspired by [Kamperis 2021] as I stated in my related work. One of these metrics is Circumference/Area (CoA). CoA measures the area and the circumference at each control point. A weight is added to each control point corresponding to the proportion of the total treatment MUs received by the control point. Hence, there exists a CoA value for an individual control point or alternatively the CoA of an entire treatment field can be expressed as the sum of the CoA over all control points.

Modulation Complexity Score (MCS) is another metric that I was introduced to through [Kamperis 2021] and further analyzed in McGiven’s original paper [McGiven et al.]. MCS calculates the leaf sequence variability (LSV) and aperture area variability (AAV) at each control point. LSV compares the distance between neighbouring leaf pairs to the largest distance between any leaf pairs on its leaf bank. AAV measures the magnitude of the area exposed by two leaf pairs relative to the largest area of those leaf pairs in the entire plan. These two scores are multiplied together along with a weight representing again the proportion of MUs received by the control point. Again, there is metric that measures the MCS for an individual control point and the MCS of the entire treatment plan is simply the sum of the MCS over all control points.

I was encouraged by Dr. Harald Keller at the Princess Margaret Hospital to implement a new metric which we called span. Span represents the distance spanned by the opening of the MLC leaves. If the collimator was not rotated then the span would simply be the distance between upper and lower jaws. Unfortunately, as the collimator rotates the calculation becomes slightly more complex. The theory behind span as a complexity metric is that as the span gets larger, more radiation hits the ArcCheck at its cylindrical parts as opposed to straight on at some of its more flat surfaces. Each control point has its own span and we have chosen to represent the span of a treatment as the maximum span of its control points.

Another metric that was implemented from Kamperis’ work [Kamperis 2021] was the leaf travel metric. Although implemented, it was not used as a predictor for the machine learning models since it is difficult to represent the leaf travel at a specific control point. This concludes the discussion of complexity metrics. Next, I introduce the machine learning for which these metrics will serve as predictors.

3.3 Machine Learning

Machine learning is used in this research to extract information from a training sample of treatment plans using statistical algorithm in order to make predictions regarding the QA pass rates of a test set of treatment plans. The

algorithms (or models) used can be classified as parametric or non-parametric. Parametric models make assumptions about the distribution of the data and have a fixed number of parameters, making them less flexible. The benefit of using parametric models is the clear interpretability of the parameters. For this reason, I have used parametric models such logistic regression and LASSO regression. These models serve as a baseline to compare the results of the non-parametric models. Further, the significance of the complexity metrics can be tested using these models. The target variable of the model is whether the treatment has passed QA (binary), hence the logistic regression. The predictors of the regression model are the plan wide complexity metrics mentioned in the previous section. The logistic regression model presents us with the coefficients associated with each predictor which can be used to determine the increase in log odds of passing the QA for each unit increase in the metric. It also produces a test statistic, whose p-value indicates the statistical significance of the complexity metric to predicting passing the QA. By analyzing the p-values of the logistic regression model, it can be determined which complexity metrics are statistically significant to passing the QA. The LASSO regression can be used to confirm these results. LASSO, a form of regularization, introduces a penalty term which limits the magnitude of the coefficients in the regression model. As the weight of the penalty term increases, the coefficients tend towards zero. The order in which coefficients become zero can be interpreted as the importance of the corresponding complexity metric for passing QA. The first coefficient that becomes zero will be associated with the least important metric and the last will be associated with the most important. The goal is to use the LASSO regression to confirm the results from the logistic regression output.

Non-parametric models are highly flexible and can capture more complex relationships in the data. Since these models do not rely on any underlying assumptions they are more flexible, leading to enhanced results in many problem spaces. Non-parametric models employed in this research include bagging, random forest, and recurrent neural networks.

Recurrent neural networks (RNNs) were selected since their feed-forward structure is suitable to measure the sequential nature of control points in a treatment plan. The recurrent neural networks are structured such that each input node corresponds to a control point. Each input node receives an array that defines that specific control point. I have built two RNN architectures in the research. The first receiving an array of the complexity metrics defining its control point discussed earlier (MCS, CoA, Span). The second receives two arrays of raw leaf tip positions. These architecture can be seen in figures 4 and 5.

Additionally, random forest and bagging models were trained and tested to provide a baseline for the performance of non-parametric methods. Although it is assumed that RNNs will perform better than these other models, they do come with some useful interpretations which are important in fields such as medical sciences (more on this in the Discussion section).

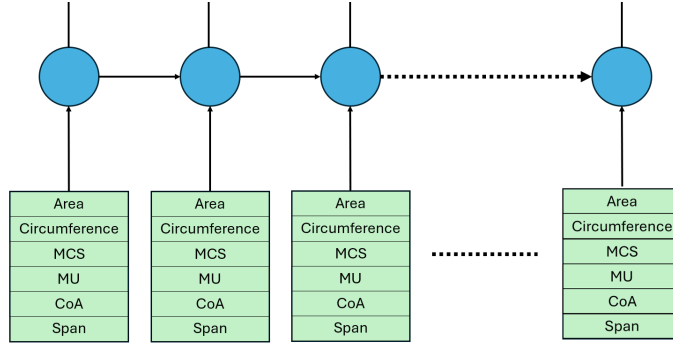


Figure 4: RNN with structured input.

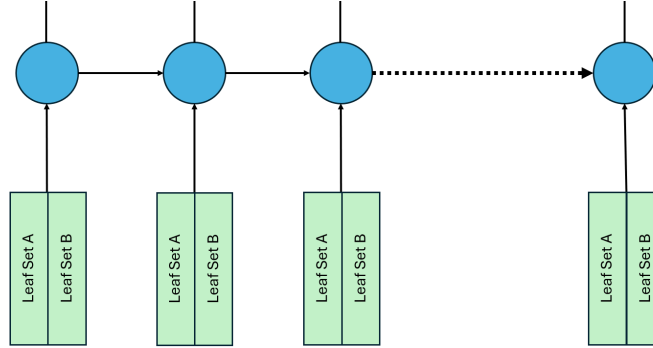


Figure 5: RNN with unstructured input.

4 Results

The models were fit using the 1464 treatment plans from Princess Margaret Hospital Elekta and Varian machines. These were split into a training set and testing set. The models were trained on the training set and hyperparameters were tuned using cross validation on the training set for random forest and bagging models. The final models were tested on the test set and their results compared. The metric used for comparing models is the area under the curve (AUC) of the receiver operating characteristic curve (ROC curve). Accuracy is also a common metric for comparing models and was used initially until it was observed that it poorly captures the true results of the model. The ROC curves and the associated AUCs can be found in Figure 6. It is clear that the RNN models outperform the rest and that the sequential structure of the data clearly needs to be accounted for when fitting VMAT machine learning models. The second RNN architecture (AUC - 0.95), which takes as input the unstructured leaf positions, performs better than the architecture which uses the calculated complexity metrics (AUC - 0.92).

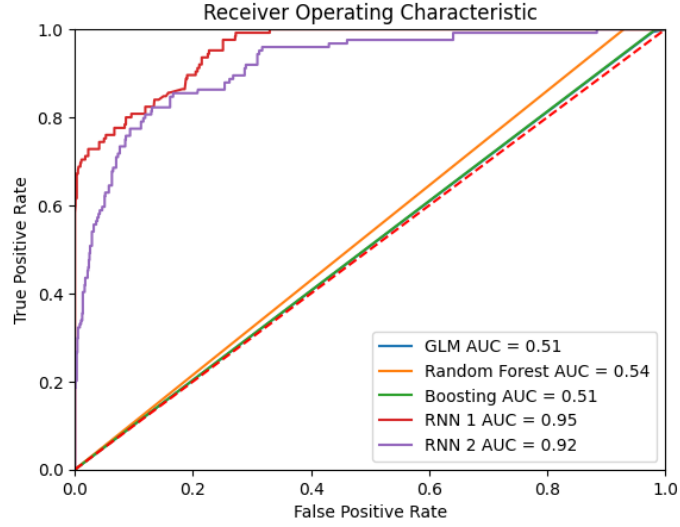


Figure 6: Receiver Operating Characteristic (ROC) curve results.

In Figure 7, the results of the p-values from the logistic regression show that at a significance level of 5%, beam intensity (field MU) is the only statistically significant complexity metrics to passing the QA. The LASSO regression also shows the coefficient of MU approaching zero last, justifying our the results from the logistic regression model.

Variable	P-value
Area	0.07675
Circumference	0.27470
Circumference over Area	0.05231
Modulation Complexity Score	0.62233
Span	0.21252
Beam Intensity	0.00226

Figure 7: P-values of complexity metrics from logistic regression model.

4.1 Discussion

With the increase in flexibility with the RNN also comes a loss of interpretability. This creates significant issues in the medical science and health care space where adopters cannot simply trust the output of a black-box machine learning model. In radiation oncology quality assurance, interpretability in machine learning is vital for ensuring the reliability and acceptance of predictive models. As these models are employed to decipher complex treatment plans data, understanding the rationale behind predictions is crucial for clinicians, researchers, and patients. Interpretability allows for the identification of relevant features and metrics implicated in the treatment field, aiding in the discovery of potential errors. Furthermore, it enables the validation of model predictions against existing medical physics knowledge, enhancing the clinical utility of these tools. Transparent and interpretable models not only facilitate the integration of AI insights into clinical decision-making but also foster collaboration between data scientists and medical professionals, fostering a more effective synergy between cutting-edge technology and traditional medical expertise in the pursuit of personalized and precise cancer treatments.

In logistic regression models, it is easy to interpret the predictions made by models. The coefficients of the model represent weight associated with each complexity metric. The prediction can be broken down and the individual impact of each predictor can be observed easily.

The interpretation of non-parametric models is significantly more challenging. For random forest models, variable importance factors can be extracted from the model. Variable importance factors quantify the extent to which individual input variables contribute to the predictive accuracy of the model by measuring the decrease in prediction accuracy when each variable is randomly permuted while keeping others constant. Figure 8 shows the variable importance factors for the complexity metrics of the random forest model. This figure captures the features that influence the overall model decision-making but lack interpretation of the prediction of individual observations. Similar to the random forest, boosting is able to represent the global importance of variables in a model. Boosting models can also represent the local importance of variables in the predictions of individual observations. Figures 9 and 10 show the global and local interpretations of the boosting model. These interpretations are significant for employing the model in quality assurance because the user can determine metrics are contributing to the prediction or patterns influencing predictions across different machines or tumor types.

Unfortunately the enhanced interpretation from the boosting model does not offset the poor model performance. Ideally, the predictions of the RNNs can be interpreted and new theory is being developed with shapley values that would do just that. Shapley values come from coalitional game theory and represent a player's contribution to the game's payout. Shapley values are incorporated into machine learning models by treating the prediction as the game, the predictors as the various players, and the payout as the difference in the specific prediction from the average prediction. For a more thorough outline of shap-

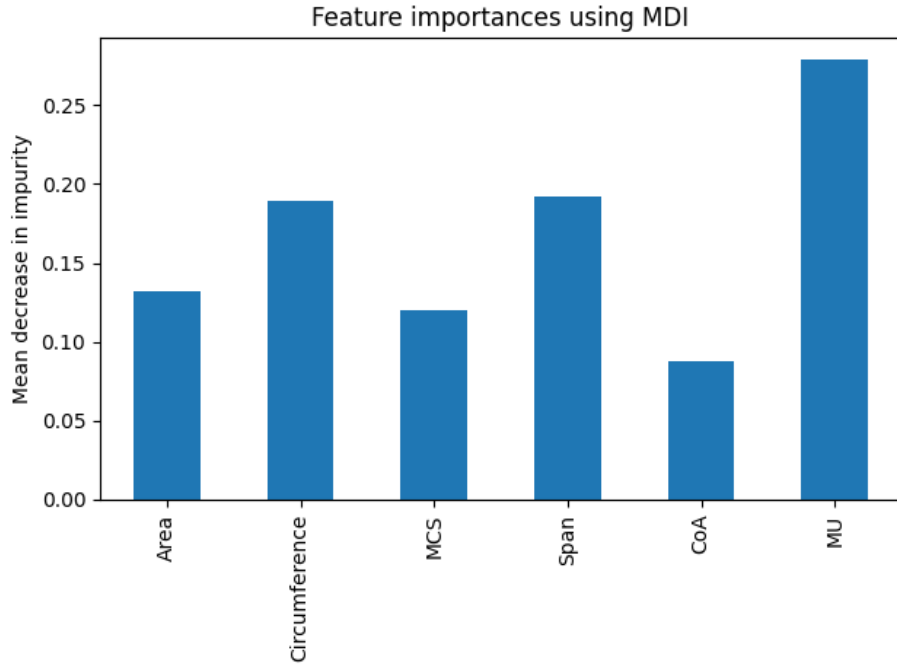


Figure 8: Random forest variable importance factors.

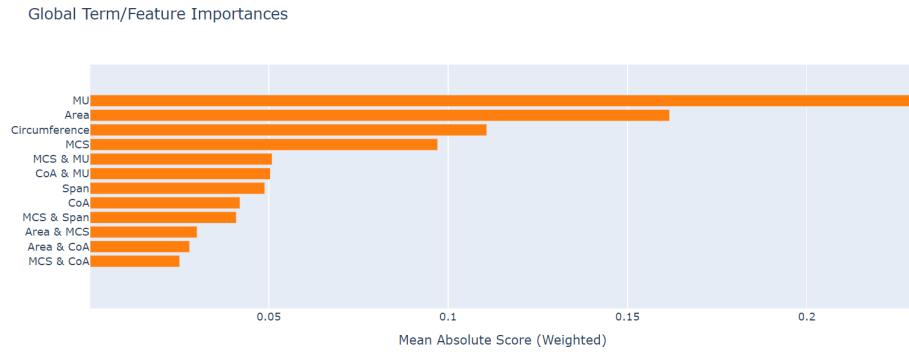


Figure 9: Boosting global variable importance.

key values and their applications to machine learning see Christoph Molnar's book [Kamperis 2021]. I attempted to implement SHapley Additive exPlanations (SHAP) interpretations of the RNN models using the shap python library but unfortunately the code was not compatible with the RNN model. The implementation of shap with the RNN models from this research would present a suitable framework for predicting the passing of QA which performs well and

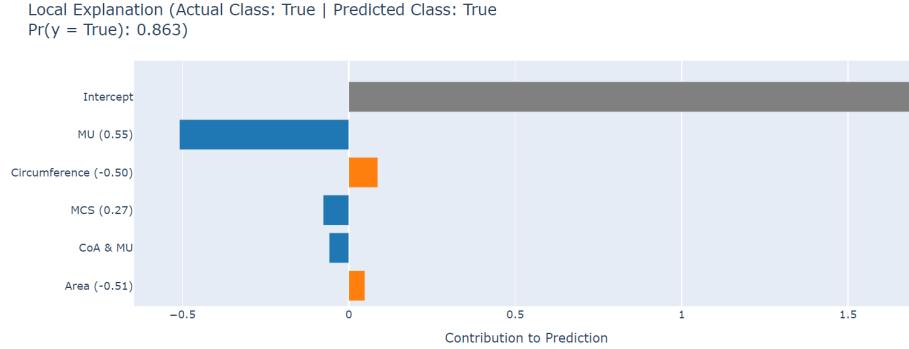


Figure 10: Boosting local variable importance.

provides accurate interpretations for end users.

5 Conclusion

In conclusion, this research addresses a critical challenge in cancer treatment by focusing on the QA process in radiation therapy, particularly in VMAT. By examining various complexity metrics derived from radiation plans, this study employs both traditional statistical methods like logistic regression and innovative machine learning techniques including recurrent neural networks (RNNs) to predict QA pass rates. The results underscore the importance of interpretable machine learning models in the clinical setting, where trust, regulatory compliance, and informed decision-making are paramount. The findings not only offer insights into the factors influencing QA outcomes but also lay the groundwork for integrating machine learning into radiation oncology practices, potentially enhancing patient care at institutions like the Princess Margaret Hospital and serving as a model for future applications in the field.

Acknowledgement

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